

Legume-based inter-cropping to achieve the crop, soil, and environmental health security

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16.1 Introduction

A gap of about 56% between the availability and demand of food will need to be filled by 2050 along with sustaining agricultural productivity, soil health, and environmental quality (Searchinger et al., 2019). The global population is projected to reach about 10 billion and the food demand will increase by over 50% by 2050. Moreover, about 800 million people are suffering from hunger and malnourishment. Hence, ensuring food and nutritional security for this exploding population will be a great challenge in the coming years (Fliebach et al., 2007). Further, the escalating land degradation enhances the emission of greenhouse gases (GHGs) and decreases the rate of carbon (C) sequestration in the soil, which contributes to climate change (Kumar et al., 2021a; Roy et al., 2021; Meena et al., 2022). Worldwide, nearly 25% of the total land area has been degraded and influencing about 3.2 billion people, particularly small and marginal farmers with poor socio-economic conditions. The intensive and unsustainable crop production practices result in the loss of fertile soils at a rate of 24 billion tons year⁻¹ and if this continues, 95% of the global land area could be degraded by the end of 2050 (GEF, 2021). In addition, the increased biotic and abiotic stresses because of climate change could be a serious threat to the food production system (Yadav et al., 2015; Kumar et al., 2016a; Prakash et al., 2020). Legumes have multidimensional action toward building nutrients pool and increasing soil fertility (Jhariya et al., 2018). The inclusion of legumes in the cropping systems contributes to providing protein-rich food and feed,

improving soil health, enhancing soil organic carbon (SOC), conserving soil and water, and improving atmospheric quality (Lal, 2017; Stagnari et al., 2017; Meena et al., 2019, 2021). The total nitrogen (N) content in the upper soil layer increased by 100 $\mu\text{g g}^{-1}$ soil in pigeon pea (*Cajanus cajan* L.)-based system as compared to 25 $\mu\text{g g}^{-1}$ soil in the cropping system without legume. Moreover, legume inclusion leads to crop diversification, which in turn reduces the incidence of harmful insect-pests, diseases, and abundance of weeds (Kumawat et al., 2020a; Lakhra et al., 2017). The production of cereals can be improved by 15%–25% by suppressing the attack of insect, pests, diseases, and weeds (Kumar et al., 2020a).

Legume inter-cropping has enormous potential to feed the ever-increasing global population through sustainable intensification under the changing climatic scenario (Iqbal et al., 2019; Jhariya et al., 2021, 2021a). It is an important agronomic practice involving the raising of two or more crops simultaneously on the same piece of land at the same time (Brooker et al., 2015; Kumar et al., 2020a). Legume inter-cropping increases the system productivity and profitability from per unit land area. Worldwide, inter-cropping supplies 15%–20% of the total food requirement (Lithourgidis et al., 2011). It creates an ecologically sound agricultural production system by reducing the use of chemical inputs which in turn produce protein, nutrients, and vitamins rich quality food (Nemecek et al., 2008). Inter-cropping vegetables, cereals, root crops, fruit plants, and other plants in the production system provides a balanced and diversified diet, boosts productivity and profitability,

and effectively utilizes resources, resulting in a sustainable agricultural system (Maitra et al., 2020; Yadav et al., 2021a,b). Hence, legume inter-cropping is ecologically, biologically, and socioeconomically beneficial in comparison with intensive sole cropping (He et al., 2012). In recent years, legume inter-cropping has driven the attention of many environmental, agricultural, and ecological researchers with extensive research interests (Brooker et al., 2015). The most of research has focused on the potential of legume inter-cropping in agricultural sustainability as they have an exceptional capacity for biological nitrogen fixation (BNF) (Stagnari et al., 2017). Inter-cropping of soybean (*Glycine max*) with maize (*Zea mays* L.) has been found beneficial and highly efficient for sustaining cropping systems for small farmers under low input and harsh environmental conditions (Iqbal et al., 2019). Legume inter-cropping can supply about 15% of the N to the component cereal crop (Li et al., 2009), which in turn increase dry matter production and reduce the external use of chemical N fertilizers and thus mitigate GHG emissions (Banik et al., 2006; Senbayram et al., 2016). Moreover, legume inter-cropping reduces the negative impact of weeds on crop productivity by suppressing their growth and biomass (Jensen et al., 2006). The wheat (*Triticum aestivum* L.) + chickpea (*Cicer arietinum*) inter-cropping reduced the weed biomass and weed population by ~70% over the sole wheat (Banik et al., 2006). Legume inter-cropping play a fundamental role in facilitating multiple ecosystem services (Jangir et al., 2017; Mitran et al., 2018; Maitra and Ray, 2019; Maitra et al., 2021), namely, (1) protein-rich quality food for human and livestock (Tharanathan and Mahadevamma, 2003); (2) recommended for low input production systems due to their ability to improve soil fertility through BNF (150 Tg N year⁻¹), enhance SOC content (0.88 Mg C ha⁻¹ year⁻¹) and mitigate GHG emissions (Kumar et al., 2018); (3) crop protection through diversification of the cropping systems (Nemecek et al., 2008). This chapter focused to introducing the legume as a inter-crop in the cropping systems to achieve the goal of agricultural long-term sustainability, food, and nutritional security under changing climate.

16.2 Legume based inter-cropping and ecosystem services

To ensuring the food and nutritional security, declining land degradation, minimizing the risk of climate change, and improving soil and environmental quality will be major challenges in the near future. Globally, about 5%–20% of human-induced GHG emissions can be limited through improved soil and crop management practices (Olsson et al., 2019). In this context, legumes could play a vital role by providing multiple agroecosystem services while following the principles of sustainability (Raj et al., 2021; Kumar et al., 2019, 2020b). It has been documented that legume crops emit 5–7 times lower GHGs per unit area than the other crops (Jeuffroy et al., 2013). Inclusion of legumes in the inter-cropping deliver multiple benefits to society and agro-ecosystem including reduce the emission of GHGs, especially N₂O and carbon dioxide (CO₂), restore soil health, control insect-pests and diseases, conserve natural resources, and reduce the use of fertilizers and energy consumption in the agricultural production system (Ghosh et al., 2007; Kakraliya et al., 2018) (Fig. 16.1). Agricultural systems (crop and animal production) release a large proportion of anthropogenic nitrous oxide (N₂O) emissions (~60%) in the atmosphere (IPCC, 2007; Reay et al., 2012). Of this, N fertilization is the main source as 1.0 kg N emits in the form of N₂O per 100 kg N applied (Jensen et al., 2012; Reay et al., 2012). The emission of N₂O is generally higher from soils covered with nonleguminous crops as compared to legume crops (Jensen et al., 2012). About 31% lower cumulative N₂O emission was observed in faba bean (*Vicia faba* L.) + wheat inter-cropping as compared to the N-fertilized sole wheat (Senbayram et al., 2016). Although, the amount of emission is governed by source and N application dose, SOC, soil pH, type of crops and cropping systems, etc. (Rochette and Janzen, 2005; Peoples et al., 2009).

The improved physiology, ecology, and agronomy under legume inter-cropping make it an advanced strategy for sustainable intensification of agricultural systems, especially under the degraded agroecosystems. Therefore, it is

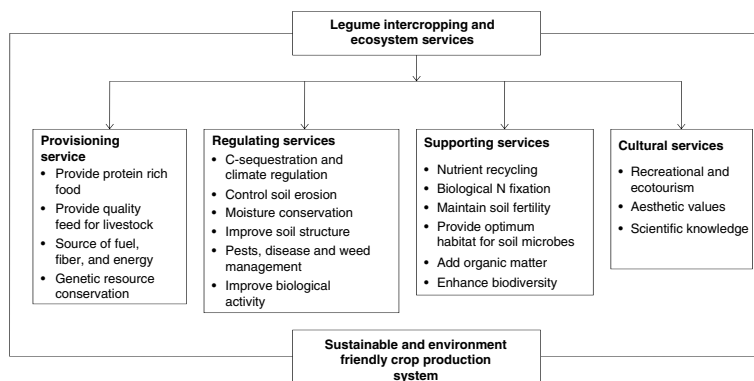


FIG. 16.1 Multifaceted ecosystem services provided by legume in the inter-cropping (Modified from Maitra et al., 2021).

necessary to implement legume inter-cropping to increase resource use efficiency (RUE), crop productivity, and to promote several ecosystem services.

16.3 Legumes as inter-crops

The key aim of inter-cropping is to produce more from per unit land area while simultaneously using the resources more efficiently and making the agricultural system more sustainable. In recent years, legume inter-cropping has been documented as an advantageous crop production system, and several legume-based inter-cropping systems are known very productive and efficient. Inter-cropping of pigeon pea with maize, sorghum (*Sorghum bicolor* L.), and soybean, potato (*Solanum tuberosum* L.) + rajmash, sugarcane (*Saccharum officinarum* L.) + lentil (*Lens esculenta* L.), linseed (*Linum usitatissimum* L.) + lentil, chickpea + mustard (*Brassica juncea* L.), etc. are potential inter-cropping systems (Ghosh et al., 2006; IIPR, 2009). Under irrigated conditions pigeon pea + groundnut (*Arachis hypogaea* L.), chickpea + mustard, chickpea + wheat/barley (*Hordeum vulgare* L.), are potential inter-cropping systems (Ali et al., 2012). Likewise, inter-cropping of spring sunflower (*Helianthus annuus* L.) with green gram (*Vigna radiata* L.) black gram has become successful in North India (Ali et al., 1998).

16.3.1 Vegetable

The present food habit is harmful to both human and terrestrial health, which needs to be shifted toward a more nutritious, and balanced diet, especially plant-based food (Mason-D'Croz et al., 2019). However, there is large gap exists between plant-based food requirements and supply, particularly vegetables, which will further increase with the increasing world population by 2050. Hence, there is a need to increase the productivity of vegetables while making efficient utilization of the resources and minimizing the cost of production. In this scenario, legume inter-cropping has the potential to reduce the production cost and enhance the agronomic yields and RUE. A farm study conducted by Yildirim and Guvenç (2005) found cauliflower (*Brassica oleracea* var. botrytis) + bean inter-cropping more remunerative, input use efficient and productive for small and resource-poor farmers over the sole cauliflower. Similarly, pea (*Pisum sativum*) + cauliflower inter-cropping produced a higher yield (22.0 t ha⁻¹), and enhanced net income (US\$ 3279), and benefit: cost (B: C) ratio (3.33) as compared to the sole crop (Anjum et al., 2015). Qasim et al. (2013) found a net benefit of US\$ 5420.2, 3527.7, and 2593.8 ha⁻¹, respectively, in pea + cauliflower, pea + garlic (*Allium sativum* L.), and pea + turnip (*Brassica rapa*) inter-cropping systems than US\$ 173.1 ha⁻¹ in pea alone. Dua et al. (2017) found a positive balance of nutrients with 36%–68%, 64%–74%, and 44%–77% of applied N, phosphorus (P),

and potassium (K), respectively, under potato + french bean inter-cropping system. Ajayi et al. (2017) found okra (*Abelmoschus esculentus* L.) + groundnut and okra + cowpea (*Vigna unguiculata* L.) inter-cropping more advantageous in terms of land productivity and monetary returns as compared to sole okra. The productivity of potatoes with dolichos (*Dolichus lablab*) and hairy vetch (*Vicia villosa* L.) inter-cropping increased by 2–3 times along with increased LER, WUE, and radiation use efficiency (Nyawade et al., 2020). The productive advantages of nonlegume vegetable + legume inter-cropping have been documented in several studies due to their complementary effects on each other, namely, tomato (*Solanum lycopersicum* L.) okra + cowpea (Olasantan, 1991), cabbage (*Brassica oleracea* var. capitata) + bean (Poniedzialek et al., 1989), watermelon (*Citrullus lanatus* L.) + soybean (Sharaiha and Hattar, 1993), chili (*Capsicum frutescens* L.) + bean (Costa and Perera, 1998), potato + lima bean (*Phaseolus lunatus*) / dolichos (Nyawade et al., 2019). Chimonyo et al. (2016) reported 46.9% higher water use efficiency (WUE) in sorghum + bottle gourd (*Lagenaria siceraria*) inter-cropping system over sole cropping. Inter-cropping of groundnut + red amaranth (*Amaranthus cruentus* L.) resulted in increased system yield, land productivity, and net returns as compared to their respective sole crops (Monim et al., 2010).

16.3.2 Cereals

The mutual benefits to both base crop and component crop in the inter-cropping system represent its potential in sustaining agricultural productivity and soil quality (Coolman and Hoyt, 1993). Legumes as an intercrop can enrich soil fertility by increasing N through BNF and other essential nutrients, while nonlegume intercrop can exhaust soil fertility. The inclusion of legumes into cereal-centric crop rotations can augment sustainability by strengthening ecological, socio-economical, and institutional aspects (Lal et al., 2016). Legume inter-cropping can also reduce the C-footprint of cereal-based systems (Gan et al., 2014). Several studies demonstrated the importance of legume inter-cropping with cereals in the tropical agroecosystem (Iqbal et al., 2019). Chimonyo et al. (2016) reported 54.7% greater WUE in sorghum + cowpea inter-cropping system over sole cropping. The land-use efficiency was enhanced by 17–53% in the inter-cropping system compared with sole crops. It has been reported that legume-based inter-cropping enhanced resource productivity and farm income by 10–14, and 52%–75%, respectively. Further, about 57% higher monetary benefits have been reported in sorghum + soybean inter-cropping systems grown in replacement series (Duvvada and Maitra, 2020). Maize is a multipurpose cereal crop and there is a wide gap between two plant rows, which offers ample scope for inter-cropping. Furthermore, the wider adaptability of maize in various seasons and agro-ecological regions makes it fit for inter-cropping. However,

TABLE 16.1 Effect of different maize + legume inter-cropping systems on system yield, soil moisture, and water and energy use efficiency (data source: Choudhary and Kumar, 2016).

Cropping system	System productivity (t ha ⁻¹)	Soil moisture contribution (mm)	WUE (kg ha ⁻¹ mm ⁻¹)	Energy productivity (kg MJ ⁻¹)	Energy output: input ratio
Maize	4.0	-25.5	4.7	0.32	10.1
Soybean	3.5	-101.9	4.8	0.43	7.0
Groundnut	4.2	-124.0	7.2	0.61	5.5
Maize + soybean (1:1)	4.5	-53.4	5.4	0.30	10.0
Maize + soybean (1:2)	5.3	-70.9	6.2	0.31	9.5
Maize + soybean (1:5)	5.6	-91.7	6.5	0.41	10.8
Maize + groundnut (1:1)	4.6	-61.6	5.6	0.30	9.3
Maize + groundnut (1:2)	5.7	-81.8	6.8	0.33	9.1
Maize + groundnut (1:5)	6.2	-103.0	7.3	0.44	9.9

the success of the system depends on the selection of crops, their planting pattern, time of planting, etc. (Maitra et al., 2019). Matusso et al. (2014) reported efficient utilization of resources along with increased yields in maize + soybean strip inter-cropping. Das et al. (2019) reported 47.4, and 43.8% higher maize equivalent yield in maize + groundnut (2:2) and maize + soybean (2:2) inter-cropping system, respectively, over sole maize. Choudhary and Kumar (2016) reported 12.5–55, and 14.9%–55.3% improved system productivity and WUE, respectively, under different inter-cropping systems over the sole cropping (Table 16.1). The energy productivity and soil moisture contribution were also higher under various inter-cropping systems.

Chen et al. (2017) observed 103.7% higher nitrogen use efficiency in maize + soybean inter-cropping study under reduced N compared with sole crop under synthetic N use.

The below-ground direct mutual advantages in cereal + legume inter-cropping systems involve increased N transfer to cereals and increased Fe and Zn bioavailability to legume crops (Xue et al., 2016; Stagnari et al., 2017). Ananthi et al. (2017) observed improved productivity, and soil fertility due to BNF (150 t ha⁻¹), reduced crop nutrients demand, and increased soil conservation in maize + legume inter-cropping than sole maize. This approach facilitates immobilization of N for long period and limits the escalating dependence on chemical N fertilizers (Regehr et al., 2015). Thus, cereal + legume inter-cropping is a sustainable agronomic strategy.

16.3.3 Legumes

Legume inter-cropping is an important strategy for sustainable intensification of agricultural systems together with

increasing net income and system productivity, reducing malnutrition, and improving ecological services (Pretty and Bharucha, 2014; Kissinger, 2016). Legumes having contrary growth and development patterns and nonsimilar plant architecture can be mutually grown in a cereal-based rotation and this technology is regarded as “doubled-up legume rotation” (DLR) (Smith et al., 2016). DLR is an alternative approach in which short stature, fast-growing, and early maturing legumes like soybean/groundnut, etc. are grown as the understory of long-lived and initially slow-growing pigeon pea. The complementary interaction in DLR reduces interspecific competition for growth resources, provides protein-dense food and fodder, maintains crop production even in a vulnerable environment, enhances soil fertility, and reduces dependency on costly fertilizers (Smith et al., 2016). Pigeon pea + groundnut inter-cropping in additive series is the most successful doubled-up legume inter-cropping system (FAO, 2016).

In India, almost all dual legume inter-cropping systems have one common intercrop component *i.e.*, pigeon pea. Pigeon pea owing to its comparatively wider row spacing, deep root system, long age, initial poor growth rate, and genotypic plasticity is the most compatible crop for inter-cropping with a wide spectrum of legumes intercrops (Saxena et al., 2018; Rani et al., 2019). In low input rain-fed production environment of Central and South India, short-duration pulses like green gram, cowpea, black gram are being cultivated with pigeon pea (Praharaj and Blaise, 2016; Saxena et al., 2018) whereas, pigeon pea + soybean is the most promising intercrops combination under high input production environments (Saxena et al., 2018). Inter-cropping of pigeon pea + soybean (2:4) enhances grain yield and net returns by 28, and 40%, respectively

(AICRPP, 2015; Rani et al., 2020; Punia et al., 2020). French bean + pigeon pea pairing in 2–3:1 was also found profitable (Rathod et al., 2004). Among various pigeon pea + legume intercrops, the highest pigeon pea equivalent yield and net returns were gained in green gram > blackgram > soybean at Maharashtra (Garud et al., 2018). Simultaneous cultivation of pigeon pea with early maturing pulses generates 4–5 q ha⁻¹ extra yields, with little-affecting pigeon pea production (Saxena et al., 2018).

Pigeon pea because of its deep root system pumps the nutrient from a deeper soil profile and later on through leaf shedding fertilizes the upper soil layers. Moreover, pigeon pea adds an enormous amount of organic matter (OM) to the soil that helps in field restoration. Soil from pigeon pea + black gram inter-cropping pattern was found rich in soil fertility parameters like available nutrient profile and SOC and also had higher dehydrogenase activity (Singh et al. 2012). Therefore, pigeon pea co-planting with legumes enhances monetary returns, makes the soil more fertile, contributes to pulse and oil production, and helps in achieving nutritional security.

Perennial forage legumes yield poorly in the first planting year therefore companion planting of annual forage legumes with perennial legumes during their crop establishment period is generally suggested. Field pea inter-cropping with red clover provided 2.56 t ha⁻¹ higher dry forage yield and reduces weeds incidence by up to 29% (Ćupina et al., 2011). Moreover, the combination of winter faba bean with winter common vetch (*Vicia sativa*), spring faba bean with spring grass pea (*Lathyrus sativus* L.), and pigeon pea with lablab bean was also found compatible in temperate regions having LER > 1. Inter-cropping of perennial forage legumes like alfalfa (*Medicago sativa* L.) and red clover (*Trifolium pretense* L.) with annuals like pea increases the first cut forage yield (Mikić et al., 2015).

16.3.4 Oilseed crops

Legume + oilseed inter-cropping provides yield and economic advantages and improves soil fertility, nutrient-use efficiency, and reduces fertilizer requirement compared with their sole crop (Dowling et al., 2021). In Brassica + clover (*Trifolium alexandrinum* L.) mixture, Brassica had higher photosynthetic activity than sole crop (Génard et al., 2017). Faba bean + lentil inter-cropping with winter oilseed rape resulted in higher growth and above ground N content and enhanced N-use efficiency, reduced weed problem, and declined stem weevil incidence (Cadoux et al., 2015). Reduced incidence of pea aphid and infection from *Mycosphaerella* fungi were also noticed in pea when grown with canola (*Brassica napus* L.) (Chalmers, 2017; WADO, 2018). The presence of canola creates a physical barrier that reduces the aphid problem in pea (Chalmers, 2017). Mixing legumes with oilseeds cut down N fertilization and improves

the nutrition of the associated oilseed (Génard et al., 2017). Oilseed canola inter-cropping with either lupine (*Lupinus* spp.), clover, or vetch increases percent N derived from the air (%Ndfa) in legume by 34%–290% than sole legume planting (Génard et al., 2017). Soil from Brassica + clover/lupine inter-cropping had higher N content during harvesting because of higher BNF by the legumes inter-crops.

Under Indian conditions, for inter-cropping with short stature oilseeds like soybean and peanut, pigeon found the best feasible option because of its complimentary growth and development compared to other legumes (Prahraj and Blaise, 2016). Green gram and black gram proved more compatible and economical with sesame (*Sesamum indicum* L.) (Awasthi et al., 2013). Moreover, the sesame-based inter-cropping systems also recorded 42%–45% higher WUE compared to sole sesame. Sunflower inter-cropping with pigeon pea, groundnut, green gram, black gram, and soybean outperformed sole cropping in terms of overall yield (Prahraj and Blaise, 2016). Because of higher monetary benefits, inter-cropping of Indian mustard + chickpea/lentil, linseed + chickpea, safflower (*Carthamus tinctorius* L.) + chickpea are recommended in northern, central, and southern regions, respectively (Prahraj and Blaise, 2016). Mustard and chickpea pairing in a 1: 2 ratio proved more productive and economical and water-use efficient under a rainfed situation (Kushwaha and De, 1987). The mixed planting also enhances nodulation and nodule dry weight than sole cropping. Groundnut being leguminous crops enhances growth characters of mustard, total yield, and economical return of the inter-cropping system when both the crops are grown in 1:1 (Wartha et al. 2020). Therefore, inter-cropping is a promising way to maintain yield levels and reduce the demand for pesticides and fertilizer, and enhances overall profitability.

16.3.5 Others

The slow growth rate along with poor crop canopy during initial development phases and wider row spacing of sugarcane crop offers great scope to accommodate short-lived crops (Geetha et al., 2018). Autumn sugarcane inter-cropping involving short-duration legumes (pea, chickpea, lentil) proved advantageous and profitable production technology that also contributes to soil health improvement (Khippal et al., 2016). Sugarcane inter-cropping with legumes improved cane yield by 15%–65% as compared to sole sugarcane (Geetha et al., 2018). Cowpea as an inter-crop in sugarcane was found effective to reduce both weed population and dry mass of weed in the initial 60 days as compared to sole sugarcane (Singh and Lal, 2008).

Sugarcane consumes a huge amount of nutrients from soil reservoirs, but its coplanting with legumes may increase soil fertility and productivity and enhances RUE. Soil pH, soil organic matter (SOM), total N, and available K were

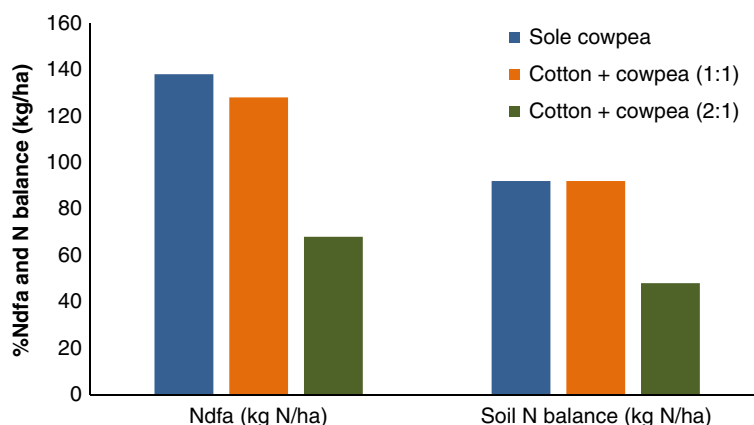


FIG. 16.2 Percent Ndfa and N balance as affected by cotton and cowpea inter-cropping (data source: Rusinamhodzi et al., 2006).

higher in sugarcane + soybean and sugarcane + groundnut inter-cropping systems (Solanki et al., 2017). Inter-cropping improves rhizospheric micro-climate and enzymatic activities and also raises N-fixing bacterial population (Solanki et al., 2017). The sole cropping of sugarcane had 42.62%, 14.5%, and 78.5% lower populations of bacteria, fungi, and actinomycetes, respectively (Li et al., 2013). Like sugarcane, cotton is also a long-duration crop with slow initial growth that provides a great opportunity to include legumes (Blaise et al., 2005). Legume-cotton (*Gossypium* spp.) inter-cropping systems make efficient resource use possible because both the crops differ in peak nutrient and moisture requirement and also extract nutrients and water from different soil layers (Sankaranarayanan et al., 2010). Bunch-type groundnut and short duration and low stature varieties of legumes are most congenial for inter-cropping with cotton. Inter-cropping of cowpea in cotton enhances cotton production efficiency by $1.9 \text{ kg ha}^{-1} \text{ day}^{-1}$ (Rajpoot et al., 2014). Cotton inter-cropping with green gram and black gram improved WUE 19.4%–22.4% compared to sole cotton (CICR, 2020). The presence of cowpea in the cotton field also reduces the sucking pest problem in cotton (Chikte et al., 2008). Cotton + pigeonpea intercrop combination, pigeon pea crop act as a trap crop for *Helicoverpa armigera* (Saxena et al., 2018). Cotton-cowpea inter-cropping attracts more numbers of beneficial pests particularly coccinellids and reported a higher rate of patronization in this system (CICR, 2020). The raised N demand in cotton + cowpea inter-cropping system resulted in higher N fixations (%Ndfa) than sole cowpea (Rusinamhodzi et al., 2006) (Fig. 16.2).

Sida hermaphrodita-a bio-energy crop when cocultivated with legume particularly *Trifolium pratense* on marginal lands resulted in 8–15 and 300% higher biomass of *Sida* and total biomass, respectively, as compared to sole crop (Moritz et al., 2018). Tobacco (*Nicotiana tabacum*) + peanut relay inter-cropping favored beneficial soil bacteria (*Actinobacteria*, *Firmicutes*, *Bacillus*, and *Lactococcus*) whereas the relative proportion of pathogenic

bacteria (*Variibacter* and *Burkholderia*) was declined (Gao et al., 2019). Rajma when intercropped with tobacco improved tobacco leaves quality in terms of higher puckering and spangling score (Singh et al., 2013).

Legumes fodder is rich in protein whereas grasses are loaded with fiber and carbohydrates, therefore excessive and sole feeding of these fodders creates bloating or dietary indigestion-related health issues in animals (Kumar and Machiwal, 2017). Moreover, cereals produce high dry matter but legumes have poor dry matter production potential (Eskandari et al., 2009). Therefore, co-cultivation of both cereal and legume-based fodder crops can provide balance quality with acceptable yield without adding any extra cost (Kumar and Machiwal, 2017). Triticale when grown with legume produced better quality forage and higher yield potential than triticale monoculture (Karadag and Buyukburc, 2004). Barley inter-cropping with lathyrus (40:60) resulted in improved forage yield and better quality (Vlachostergios et al., 2015).

16.4 Legumes inter-cropping and soil health

Legumes play a crucial role in restoring soil health and offer a potential strategy for agricultural and ecological sustainability. Residues of legume crops increase microbial abundance and their functions by providing energy and optimum soil environment, adding a large amount of OM to the soil, controlling soil, and wind erosion, improve soil structure and water retention capacity, etc. (Lal, 2011). Thus, legume residue acts as a soil conditioner. Legumes serve an important service to the agroecosystem by converting atmospheric inert N into plant-available form through BNF in the association of symbiotic bacteria and improve soil health (Lal, 2017). Including legume crops in the cropping systems is an integral component of integrated nutrient management in developing nations (Sharma et al., 2019; Kakraliya et al., 2017, 2018; Kumar et al., 2022). Legume inter-cropping

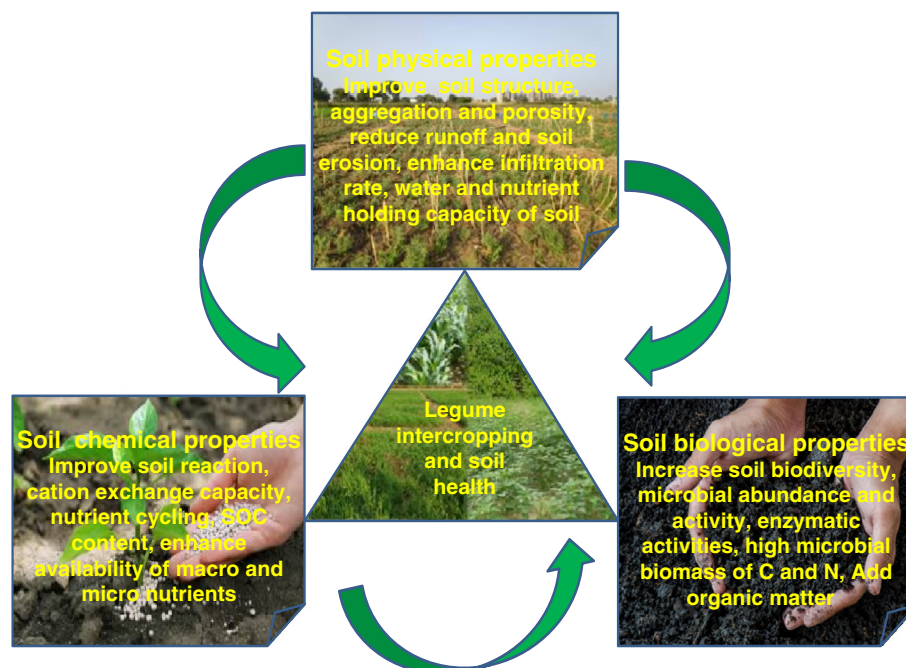


FIG. 16.3 Positive impact of legume inter-cropping on soil health parameters.

improves soil's physical, chemical, and biological properties, which are interrelated with each other (Meena et al., 2020,a,b) (Fig. 16.3). Pulses and oilseed legumes contribute up to 2.95 and 18.5 Tg N year⁻¹, respectively, through the process of BNF (Herridge et al., 2008). About 1.0 kg N ha⁻¹ day⁻¹ can be fixed by legumes (Singh and Chahal, 2020). This helps in reducing the requirement of inorganic N fertilizers in the associated crop and succeeding crops.

Besides, the BNF process enhances the mycorrhizal colonization and functions of soil microorganisms (Nasto et al., 2014). The increased activity of symbiotic microbes helps in mitigating the chronic P deficiency of several tropical ecosystems by mineralizing organic P (Ghosh et al., 2007). The increased microbial activity also helps in mineralizing other essential plant nutrients into a more soluble form, particularly rhizospheric N (Phillips et al., 2013). This increased activity enhances the use of complementary resources among legumes and associated crops, which promotes the abundance, functions, and diversity of microbes (Ashton et al., 2010; Xu et al., 2019). The improved N content in the soil also increases SOC and overall soil health. In this context legume inter-cropping has been identified as a sustainable agronomic approach to contribute N and create low N input agroecosystems (Fustec et al., 2010).

Inter-cropping of maize with soybean resulted in improved soil physical, chemical, and biological properties (Oelbermann et al., 2015). They found 8.9%–16.9% and 1.5.2%–31.0% higher bulk density (BD) at both 0–20 and 20–40 cm soil depths, respectively. The SOC, N, C: N ratio, and C stocks were significantly higher in 2012 at

both depths. Similarly, 23.5%, 41.7%, and 14.6% at 0–10, 10–20, and 20–40 cm depths increased the microbial biomass carbon (MBC), respectively, in the inter-cropping system over sole maize in 2012. The microbial biomass nitrogen (MBN) was also increased by 8.7% at 0–10 cm depth under the inter-cropping system than the maize alone in 2012 (Table 16.2). In this way, legume inter-cropping is crucial for adaptation and mitigation to climate change, agricultural sustainability, restoring soil health, and making the food production system more profitable (Maitra and Ray, 2019).

16.4.1 Soil physical

Soil erosion, nutrient mining, and depletion of SOC are major factors responsible for land degradation and deterioration of soil physical properties, which consequently results in low agronomic yields of agricultural systems. Further, the input-intensive cropping systems accelerate the degradation of soil fertility and perpetuate nutritional insecurity (Lal, 2017). In this scenario, legume inter-cropping is an advanced approach for improving soil physical properties and sustaining agricultural production (Begam et al., 2020). Gong et al. (2019) noted significantly lower BD and soil temperature under prosomillet (*Panicum miliaceum* L.) + green gram inter-cropping pattern in loam texture of Loess Plateau of China. The increased root biomass and activities and addition of crop residues into the soil in the legume + maize inter-cropping system have resulted in improved aggregate stability and mean weight diameter and

TABLE 16.2 Changes in soil health parameters under maize + soybean inter-cropping at different soil depth (data source: [Oelbermann et al., 2015](#)).

Soil health parameters	Soil depth (cm)	Sole maize		Maize + soybean(1:2)	
		2007	2012	2007	2012
BD (Mg m^{-3})	0-20	1.23	1.34	1.18	1.38
	20-40	1.25	1.44	1.16	1.52
pH	0-20	5.9	5.7	5.9	5.6
	20-40	6.1	5.9	6.1	5.8
SOC (%)	0-20	3.57	4.52	3.35	4.48
	20-40	1.74	2.44	1.79	2.69
N (%)	0-20	0.21	0.22	0.20	0.23
	20-40	0.15	0.18	0.15	0.20
C: N ratio	0-20	16.7	20.3	17.4	19.7
	20-40	12.0	13.6	12.0	13.5
SOC stock (g m^{-2})	0-20	6191	8065	5413	8432
	20-40	4453	3531	4172	3824
MBC ($\mu\text{g g}^{-1}$)	0-10	243.8	308.0	259.8	380.3
	10-20	249.1	214.4	182.1	303.7
	20-40	138.5	172.7	178.6	197.9
MBN ($\mu\text{g g}^{-1}$)	0-10	41.4	31.7	37.8	60.0
	10-20	41.3	27.1	31.4	28.0
	20-40	53.7	24.3	23.4	20.3

reduced BD and penetration resistance of soil ([Latif et al., 1992](#)). Potato intercropped with beans, peas, and dolichos resulted in 7%, 9%, and 13% higher soil moisture content, respectively, across the seasons and slope positions. The addition of higher OM and extensive root system of sorghum + lablab inter-cropping systems led to improved hydraulic conductivity, aggregates stability, total porosity, and stability index as compared to the respective sole cropping systems ([Mathan, 1989](#)). Legume inter-cropping is beneficial in reducing erosion losses by providing more dense cover against striking action of rainfall on the ground surface ([Maitra et al., 2020](#); [Kumawat et al., 2020b](#)). The Deep tap-roots of legume break the hardpan and utilize the nutrients and moisture from deeper layers of soil, while the shallow roots near the soil surface reduce soil erosion by binding the soil particles ([Lithourgidis et al., 2011](#)). [Sharma et al. \(2017\)](#) reported 26, and 43% reduced runoff and soil loss, respectively, and also 13% higher productivity of succeeding crop under inter-cropping system due to improved soil structure and moisture content. [Singh et al. \(2020\)](#) also observed 24% reduced soil loss and 17.5% reduced runoff

in the inter-cropping system compared to sole crop. A field study conducted in deep Vertisols by [Singh et al. \(2014\)](#) reported that dense canopy cover under soybean + maize and maize + pigeon pea inter-cropping systems reduced the runoff by 23.7, and 19.3%, respectively, over the sole maize crop. Reduction in soil loss was 34.5, and 24.4% in soybean + maize and maize + pigeon pea, respectively, over the sole maize crop. The Mean cumulative sediment yield decreased from 169 t ha^{-1} in sole potato plots to 50–83 t ha^{-1} in potato-legume inter-cropping, representing a reduction of 51%–70% ([Nyawade et al., 2019](#)).

16.4.2 Soil chemicals

Legume inter-cropping has a positive impact on soil chemical properties. Soil chemical parameters, namely, pH, cation exchange capacity (CEC), nutrient content, and SOC concentration are primarily used for assessing soil fertility and sustainability of the agroecosystem ([Yuvraj et al., 2020](#)). Including legume crops in the cropping system alters the pH of the rhizosphere, which increases the activity of

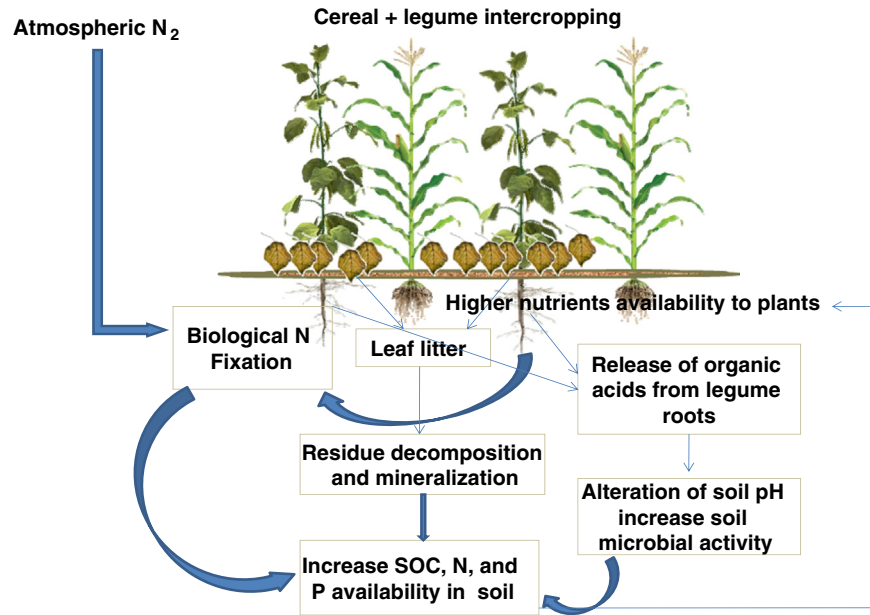


FIG. 16.4 Influence of legume inter-cropping on soil chemical properties.

soil microorganisms and thus regulates the process of nutrient cycling (Fig. 16.4) (Oelbermann et al., 2015). The secretions of organic acids from crop roots and CO_2 upon decomposition of residues under the legume inter-cropping system decrease the pH of alkaline soils (Mitran et al., 2018). Integrating legumes as inter-crop in the cropping systems helps in buffering soil pH. Inter-cropping of green gram with prosomillet significantly lowered the soil pH and increased SOC, and total N, P, and K over the pure stand of pearl millet (*Pennisetum glaucum* L.) (Gong et al., 2019). Similarly, soil available N, $\text{NH}_4^+\text{-N}$, $\text{NO}_3^-\text{-N}$, P, and K were higher under inter-cropping systems (Li et al., 2013, 2016; Gong et al., 2019).

SOC is crucial for soil health and including legumes as inter-crop in the cropping system has great potential to enhance SOC content into the soil and thus, improve soil health (Lal, 2016). Legume + cereal inter-cropping enhances yields per unit area and may add higher C-biomass into the soil over the sole cropping (Lithourgidis et al., 2011). Maize + soybean inter-cropping in 2: 3 ratio recorded 23.6 g C kg⁻¹ of soil as compared to 21.8 g C kg⁻¹ of soil in sole maize, showing the potential of legume inter-cropping for increasing SOC content (Bichel et al., 2016). Integration of legumes with cereals increases soil fertility by contributing 80–350 kg N ha⁻¹, which reduces the loss of N by improving N uptake (Mobasser et al., 2014). Maize intercropped with legume crop enhanced the N uptake by 47.02% and reduced the dose of N application from 240 kg N ha⁻¹ to 180 kg N ha⁻¹ (Shaoming et al., 2004; Yang et al., 2014). Gitari et al. (2019) noted that inter-cropping of potato with pea, dolichos, and beans improved SOC by 20, 42, and 21%, respectively, along with improved soil pH, and available

N, P, and K content. Similarly, the CEC was increased by 18.3, 4.7, and 13.6% under potato inter-cropped with dolichos, pea, and beans, respectively, over the sole potato. Maize + peanut inter-cropping has been reported to increase the nutrient (N and P) supply and soil microbial diversity, which plays a crucial role in improving plant health and soil fertility (Li et al., 2018; Kumar et al., 2021a).

Next to N, P is an important nutrient for plants and indispensable for energy transfer within plant cells. Legume crops increase the availability of P by excretion of organic acids, release of phosphatase enzymes, and reducing soil pH in the rhizosphere (Varma et al., 2017). Although, the efficiency of organic acids of P mobilization largely depends on the type of soil and soil environmental conditions (Hinsinger, 2001). Lupin excretes both organic acids as well as phosphatase enzymes (Gilbert et al., 1999). Gardner and Boundy (1983) noted increased uptake of P by wheat intercropped with lupine because of increased P dissolution and availability in the calcareous soil. The deep and intensive root system of legume crops also recovers P from insoluble P (Lynch, 2007). Faba bean is one of the major legume crops to recover P from nonavailable form (Nuruzzaman et al., 2005). The acidification of rhizosphere enhanced the P availability in the maize + cowpea inter-cropping system than the sole cropping (Latati et al., 2016). Wang et al. (2012) found the accumulation of NaOH extractable P pools and acid-extractable P pools in the rhizosphere of legume inter-cropping. The extensive root system and altered microbial composition in cereal + legume inter-cropping systems lead to the increased availability of P, Fe, and Zn in the rhizosphere and thereby increased uptake by the crops

(Xue et al., 2016; Maitra and Ray, 2019). Hence, legume inter-cropping improves soil chemical properties such as pH, CEC, SOC, total and available N, and available P, and K content (Jensen et al., 2012; Gitari et al., 2019).

16.4.3 Soil microbes

Soil microorganisms are a vital component of the planet's biodiversity and play a crucial role in the decomposition of OM, nutrient recycling, N fixation, and improvement of soil structure (Ludwig et al., 2015; Gong et al., 2019). The inclusion of legumes as intercrop increases the diversity of soil flora and fauna resulting in more stability to the soil life (Begam et al., 2020; Maitra and Ray, 2019). The legume inter-cropping modifies the root distribution pattern and their structure and exudation of organic acids in the rhizosphere by developing the N fixing capability of roots. These modifications have a positive impact on soil micro-organisms and their interaction with crops (Singh and Chahal, 2020). Legume crops also accelerate microbial biomass in the soil by supplying extra N through BNF. Inter-cropping of sugarcane with green gram, cowpea, and lentil increased the MBC by 8.1, 19.7, and 12.6%, respectively. Similarly, the soil MBN improved by 60.5%, 52.4%, and 75.8%, respectively, in sugarcane intercropped with green gram, cowpea, and lentil over sugarcane alone (Suman et al., 2006). The greater diversity of rhizo-deposits in the inter-cropping of durum wheat with chickpea/lentil also led to increased MBC and microbial biomass phosphorus in the rhizosphere over the sole cropping (Tang et al., 2014). The increased interaction of soil microbes with crops in the rhizosphere under legume inter-cropping increased the agronomic yields by inducing plant growth and development (Li et al., 2013). The BNF process of legume crops enhances the biological functions of soil and moderates the release of N_2O into the atmosphere (Peoples et al., 2009). Solanki et al. (2017) found increased diazotrophs population in plant rhizosphere and endosphere, dehydrogenase enzyme activity, and overall biological activity in sugarcane + legume inter-cropping over their solitary cropping. Li et al. (2013a) observed that inter-cropping of soybean with sugarcane significantly increased the population of bacteria, fungi, and actinomycetes by 43.9–188.0, 14.6–72.2, and 41.1–73–4%, respectively. Legume inter-cropping with soybean also led to significant improvement in the activities of nitroreductase, urease, and acid phosphatase enzymes in the soil (Li et al., 2016). The improved abundance of N-fixing microorganisms in the soil rhizosphere has been reported in groundnut + maize inter-cropping (Chen et al., 2018). Gong et al. (2019) reported that inter-cropping of green gram with prosomillet significantly increased the activities of soil invertase, urease, and catalase enzymes by 18.4%, 29.1%, and 42.9%, respectively, over the sole cropping (Li et al., 2016).

16.5 Legumes inter-cropping and plant health

Plants are sessile, have a greater chance to interact with their immediate environment (Latef et al., 2016). Plants are often exposed to diverse biotic (insect-pests, diseases, weeds, etc.) and abiotic (light, temperature, water, nutrients, etc.) stresses which affect their growth and limit agricultural productivity (Ramegowda and Kumar, 2015). Further, evidence advocate that the changing climate will expand the chances of biotic and abiotic stress occurrence in plants (Garrett et al., 2006). About 70% reduction in crop yield has been recorded due to abiotic stresses (Kumar et al., 2013). According to Food and Agriculture Organization (FAO) estimates, about 40% of global crops are lost annually due to the incidence of various types of insect-pests and diseases. In this context, the United Nations has also declared 2020 as the International Year of Plant Health to aware people of the role of plants in food security, poverty reduction, environment protection, and advancement of economic development. Hence, the adoption of an ecosystem approach involving the integration of various management strategies is needed to minimize the use of pesticides, and avoid the adverse impact of biotic and abiotic factors on plant health. Legume inter-cropping is a potential agronomic strategy for improving agronomic yields, environmental quality, and plant health by using light, water, land, and nutrients more efficiently (Brooker et al., 2015). Moreover, inter-cropping also reduces the incidence of insect-pest and diseases and suppresses the weed biomass by altering the physical structure of plants or manipulating crop microclimate (Coolman and Hoyt, 1993). Legume inter-cropping not only avoids the use of chemical inputs but also protects natural enemies of pests, pollinators, and beneficial organisms and replenishes soil fertility (Fig. 16.5).

Certain basic nutrients are required for plant growth and health (Maitra et al., 2020). Legume crops can fix atmospheric N through BNF where rhizobium bacteria live in nodules developed on plant roots and convert them into plant absorbable form (Lal, 2017). Further, the importance of BNF by legume crop increased in the N-limited soils. The N content and absorption were increased in maize when intercropped with pigeon pea due to increased supply of N through BNF by pigeon pea (Seran and Brintha, 2010). Xue et al. (2016) reported increased productivity, grain quality, and plant health due to increased availability and uptake of P, Fe, and Zn under cereal + legume inter-cropping system. The improved field microclimatic conditions under maize + soybean inter-cropping improved the photosynthetic rate and biological parameters and thus enhanced the system productivity (He et al., 2012; Iqbal et al., 2019). Legume inter-cropping improves the grain quality of cereals, especially protein content in the cereal + legume inter-cropping system

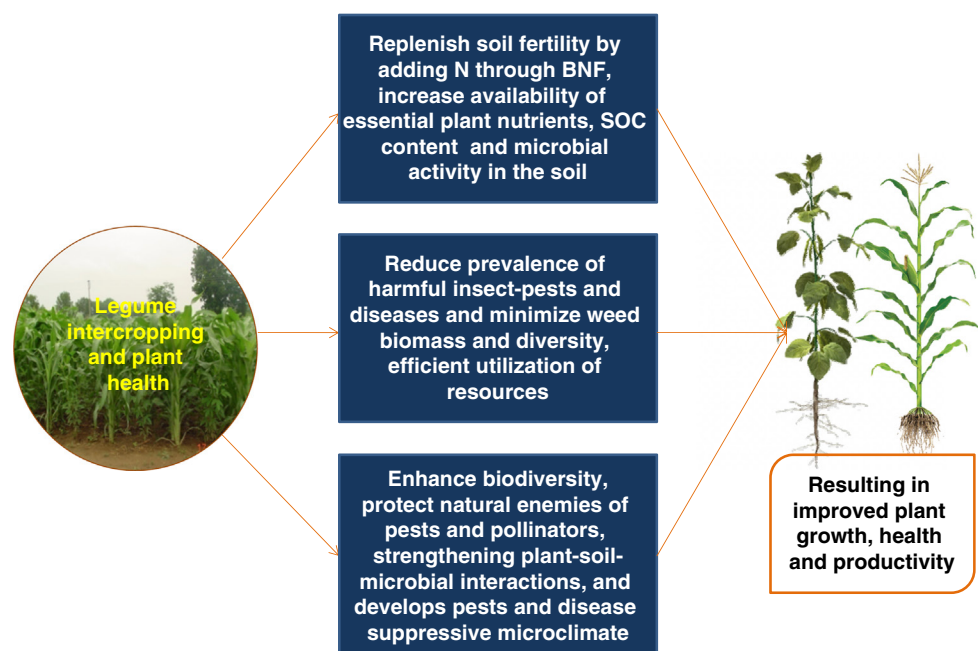


FIG. 16.5 Affirmative role of legume inter-cropping in improving plant health.

(Gooding et al., 2007). The positive interaction of plants-soil-microorganisms under legume inter-cropping can play an important role in enhancing the availability and effective use of nutrients and plant health (Tong et al., 2015). Thus, legume inter-cropping is one of the most sustainable ecosystem approaches for maintaining plants' health by providing favorable microclimate and replenishing soil nutrients and thus, helping in achieving SDGs of food and nutritional security and zero hunger.

16.6 Legume inter-cropping for insect-disease and pest management of component crops

Insect-pests, diseases, and weeds are of serious concern for the food production system, especially in intensive and mono-cropping systems (Siddique et al., 2012). The distribution and severity of pest outbreaks may be further accelerated by climate change, which may affect both the quality and quantity of agricultural products (Kumar et al., 2017,a,b). Legume inter-cropping could be a sustainable way for minimizing the prevalence of insect-pests, and diseases and suppressing weed biomass (Table 16.3). Diversification of cropping systems through legume inter-cropping increases the abundance of beneficial pest parasites and predators and maintains the harmful pests' population below the economic threshold level (Maitra et al., 2019). Further, legume inter-cropping strongly follows the concept of "push and pull," which is an effective strategy for insect-pests and weed control (Hassanali et al., 2008; Xu et al., 2017). Hence, legume inter-cropping is

an ecosystem-based approach, which reduces the chemical pollution in the agricultural system by reducing the use of chemical pesticides (Singh and Adjeigbe, 2002). Legume inter-cropping decreases the incidence and multiplication of disease-causing pathogens by increasing the functional diversity in the system (Kumawat et al., 2020a). Moreover, the legume-associated *Rhizobium* species in the inter-cropping systems hinder the growth of soil-borne pathogens and develop a complex environment that restricts the reproductive functions of pathogen, namely, *Fusarium* spp., *Phytophthora* spp., etc. (Maitra and Ray, 2019). Maize + legume inter-cropping was found superior in reducing the population of harmful pests of maize, namely, corn borer, budworm, stalk borer, and leafhopper, and increasing population of beneficial insects (Seran and Brintha, 2010). In contrast, inter-cropping of maize with legume has been reported to limit the severity of some legume diseases such as Ascochyta blight and angular leaf spot than sole legume crop (Schoeny et al., 2010). Various experiments were carried to assess the yield decline and extent of harm caused by stem borer in maize + soybean, maize + cowpea, and sole maize. Results showed 21.3%–48.1% increased density of larva and 1.8–3.0 times higher yield loss in sole maize than inter-cropping. Moreover, increased stem tunnels and damaged cobs were found in sole maize (Chabi-Olaye et al., 2005). Sekamatte et al. (2003) reported reduced attack of termites in maize + soybean inter-cropping. Legume inter-cropping boosts the maize yield by improving soil health and avoiding the attack of problematic pest stem borer from the crop (Ziv et al., 2012).

TABLE 16.3 Role of legume inter-cropping in insect-pests and disease management.

Inter-croppingsystem	Insect-pests and disease management	References
Pearlmillet + groundnut	Infestation on pearl millet reduced by 30.6%–47.9% and the population of stem borers per plant reduced by 33.7%–58.2%	Degri et al.(2014)
Sorghum + cowpea	Lowered the numbers of stem borer, spotted stem borer in sorghum, and thrips in cowpea. The yield of unprotected intercropped sorghum was reduced by 15% while the yield of unprotected sole sorghum was reduced by 28%	Ampong-Nyarko et al. (1994)
Wheat + pea	Reduced aphid abundance and plant infestation in wheat	Xu et al. (2017)
Sorghum + cowpea/ groundnut/pigeon pea	Reduced stem borer damage in sorghum and recorded 10%–38% incremental benefits over sole cropping	Muturik et al. (2018)
Wheat/barley + faba bean	The incidence of yellow rust (<i>Puccinia striiformis</i> f.sp. <i>tritici</i>) and mildew (<i>Blumeriagraminis</i>) in wheat/barley, and chocolate spot (<i>Botrytis fabae</i>) and fusarium wilt (<i>Fusarium oxysporum</i>) in faba bean reduced by 45% on average	Zhang et al. (2019)
Tomato + cowpea	Control of bacterial wilt disease of tomato	Michell et al. (1997)
Barley + lupin/faba bean/pea	Reduced incidence of net blotch (<i>Pyrenophora teres</i>), brown rust (<i>Puccinia recondita</i>), and powdery mildew (<i>Blumeriagraminis</i> f. sp. <i>hordei</i>) in barley and Ascochyta blight (<i>Ascochyta pisi</i>) in pea	Kinane and Lyngkjær (2002)

Legume inter-cropping also plays a vital role in weed management by providing excellent ground cover, which helps in suppressing weed growth. The introduction of legumes as an intercrop with maize has been reported to limit weed biomass and density than the sole maize (Bilalis et al., 2010). Legume inter-cropping creates both spatially and temporally competitive environments against weeds in the fields. Low weed biomass, weed density, weed diversity, and weed infestation were observed in maize + soybean inter-cropping systems than the respective sole crops, which resulted in increased productivity of the system (Kumar et al., 2003; Matusso et al., 2014). Similarly, Choudhary and Choudhury (2018) documented lower weed density and biomass and higher weed smothering efficiency in maize + legume inter-cropping than the pure stand of maize. However, the efficiency of legume inter-cropping in suppressing weed growth highly depends on the type of component crops, spatial arrangement, plant density, soil moisture content, and fertility status of soil (Belel et al., 2014).

The lower NPK removal through weeds in maize + soybean and maize + cowpea by 37.4, and 38%, respectively, increased the system productivity (Chalka and Nepalia, 2006). The reduced availability of light and space in maize + soybean inter-cropping increased the weed smothering efficiency and thus, reduced the weed biomass (Bilalis et al., 2010). Thus, legume inter-cropping provides insect-pests, disease, and weed suppressive environment for crop production and reduce the use of chemical pesticides polluting agroecosystem together with improved soil health (Gronle et al., 2015; Begam et al., 2020).

16.7 Constraints and challenges for legumes in inter-cropping system

Although, legume inter-cropping serves many significant benefits to the agroecosystem but there are some constraints and challenges also which limit the large-scale adoption of the inter-cropping system in modern agriculture (Stagnari et al., 2017; Layek et al., 2018). These constraints and challenges may be socio-economical, agronomical, institutional, and environmental. The aforementioned constraints need to be focused to popularize the legume-based inter-cropping systems in a large area (Bationo et al., 2011; Maphumo et al., 2011). The following are major constraints and challenges limiting the large-scale adoption of legume inter-cropping (Lal, 2017; Stagnari et al., 2017; Layek et al., 2018; Singh and Chahal, 2020).

- The limited and nonavailability of quality seeds, synthetic fertilizers, biofertilizers, and other farm inputs, lack of technical support, and scientific knowledge about multifaceted inter-cropping systems in the developing countries leads to the poor performance of the system (Jangir et al., 2016; Layek et al., 2018).
- Nonavailability of suitable farm machinery along with the high cost of purchasing.
- Poor market facilities and lack of financial support, infrastructures, and government policies restrict the popularization of inter-cropping systems (Maphumo et al., 2011).
- A large gap exists between the technology developed by the scientists at research farms with all facilities and

technology adopted by resource-poor farmers (Matusso et al., 2014; Kumar et al., 2016a).

- Lack of well-developed extension services and constraints related to agronomic management, especially in mechanized and intensive crop production systems.
- Sometimes, legume inter-cropping reduced the yield of the base crop than their pure stand yields, which limits the popularization of legume inter-cropping.
- The dense canopy cover in legume inter-cropping may develop a microclimate that may be favorable for the growth and multiplication of fungal pathogens (Boudreau, 2013).
- The demand for a single product by the consumers, and easy access to suitable machinery and other agricultural inputs for intensive crop production systems. Complexities of insect pests, disease, and weed infestation, socio-economic status of crop producers, changing climatic conditions, increasing vulnerability of cropping systems to climate change, poor storage facilities, etc. (Singh and Chahal, 2020).
- The sensitivity of legume crops toward soil salinity and alkalinity conditions, especially in arid and semiarid regions (Singh and Chahal, 2020).
- The lower availability of P and some microelements in acidic soils lower the crop productivity (Sanchez et al., 1997; Layek et al., 2018).

16.8 Promotional opportunities for legumes in inter-cropping system

Introducing legume crops in inter-cropping systems has great potential for sustainable intensification of agriculture using agricultural resources more efficiently, satisfying rising human needs together with improving the environment and soil quality (Dwivedi et al., 2016; Ghosh et al., 2016). The promotion of legumes as inter-crop in the cropping systems offers vital opportunities for ensuring food and nutritional security at regional and global level, enhancing the livelihood of small and marginal farmers, and sustaining soil health (Ghosh et al., 2007; Lal, 2017; Layek et al., 2018). Legume crops can fit into various cropping systems and provide an opportunity to boost the agronomic yields of inter-cropping systems by improving soil fertility (Ghosh, 2004).

The intensive agricultural practices degrade soil fertility, accelerate soil erosion, deplete SOC, and cause the loss of biodiversity (Ladha et al., 2003). Conservation agriculture (CA) is a sustainable crop production system to overcome the challenge of intensive agricultural practices and mitigate the adverse impact of climate change (Kassam et al., 2019; Kumawat et al., 2020c). Legume crops do not require fine seedbeds and can perform well under rough seedbed conditions. Hence, CA has good potential for introducing legume crops as inter-crops in the cropping systems. ZT

with residue retention improved the yield of green gram than conventional tillage (CT) practices (Kumar et al., 2012). This may be due to increased microbial abundance and their activities and improved SOC and N in the soil (Kumawat et al., 2021). Yadav et al. (2021) reported that retention of soybean residues under CA results in improved microbial activities, SOC, available N, P, and K, moisture content, and soil porosity in Vertisols of Central India. Land surface configuration practices, namely, raised and sunken beds and modified furrow-irrigated raised beds can provide great opportunity to promote legumes as intercrop by sowing them on raised beds with water-demanding crops, such as rice in sunken beds (Das et al., 2014). Bed-planting techniques offer immense potential for legume inter-cropping, especially in wide-spaced crops. In Indian Himalayan Region, about 10% of the agricultural land is used for field bunding and other conservation measures due to slopping lands and erosion problems. Legume crops like soybean, pigeon pea, black gram, green gram, etc., can be cultivated to increase production, conserve the natural resource, and enhance net income (Das et al., 2016). Crops with wider row spacing such as maize, sugarcane, cotton, etc., are also important for promoting legume crops as an intercrop (Gupta et al., 2002). Cultivating two lines of French bean (30 cm spacing) as intercrop in between wider maize rows (60 cm spacing) reduced the cost of stacking by 50% along with increased moisture conservation, time saving, and improved soil quality (Kumar et al., 2018). Inter-cropping of short-duration green gram and black gram with spring-planted sugarcane is a potential system under irrigated conditions (Panwar et al., 1990). Pearl millet is an important crop of subtropical drylands offering good scope for introducing grain legumes in the production system (Ghosh et al., 2007; Ram and Meena, 2014).

In South Asia, nearly 22.2 million hectares of area are left fallow after harvesting rice due to various crop production-associated constraints (Gumma et al., 2016). Of this fallow land, 88% is available in India alone showing ample scope for grain legume introduction as relay inter-cropping (Kumar et al., 2018). Legumes with the quality of short duration, low resource requirements, and restoration of soil health are the best crops for enhancing productivity and profitability of rice-fallow agroecosystems and providing livelihood to millions of resource-poor small farmers (Das et al., 2018). Small-seeded legume crops such as lathyrus, lentil, etc., can be grown as relay inter-cropping also known as *utera* or *paira* cropping in rainfed lowland rice-fallow areas of North-East Region of India (Das et al., 2014). Das et al. (2014) reported 400–600 kg ha⁻¹ seed yield of lentil under *utera* cropping in the North Eastern Himalayan region of India. However, the success of relay inter-cropping depends on the selection of crops and their varieties, resource conservation practices, and agronomic management practices (Kumar et al., 2016b, c). The inclusion of

legume crops in shifting cultivation offers enormous potential to promote legume-based inter-cropping for enhancing the productivity and livelihood of tribal communities. The inter-cropping of grain legumes with cereals in the terrace and shifting cultivation is also a possible way to popularize legume inter-cropping. In Nagaland, India, grain legumes like soybean, rice bean (*Vigna umbellata* L.), pigeon pea, etc. are intercropped with rice, maize, and millets under shifting cultivation (Das et al., 2016). Legume crops are also cultivated with forest trees (alder-*Alnus nepalensis*) as an agroforestry system under shifting cultivation in the Indian Himalayan Region (Kumar et al., 2018; Meena et al., 2018).

16.9 Possible future policy and action plan

The multifaceted benefits of legume inter-cropping including enhancing agronomic yields and net income, sustaining soil and plant health, minimizing risks associated with insect-pests, diseases, and weeds, and increasing RUE have been well documented. This legume inclusive system has immense potential for ensuring food and nutritional security and developing sustainable agro ecosystems under changing climatic conditions. However, there is a need to develop and identify the region-specific legume and component crops for the success of legume inter-cropping systems. The quantification of the N application rate in different inter-cropping systems is essential for higher nutrient utilization efficiency and profitability. Global popularization of legume inter-cropping through field demonstrations, dissemination of scientific knowledge, and large-scale extension activities is the need of the hour for livelihood security of resource-poor small landholders and soil and agricultural sustainability. There is a need for knowledge dissemination about the positive impact of legume inter-cropping on soil and environmental security in the long term. Understanding and developing location-specific agronomic management practices is in increasing demand for the success of inter-cropping systems. Legume inter-cropping acts as a soil amendment for improving soil physical, chemical, and biological health which need to be focused on to achieve the UN's sustainability goals. Research needs to be focused on the effect of legume inter-cropping on system productivity, soil sustainability, and soil microbial and insect-pest dynamics under diverse agroecosystems. CA is an advanced approach to sustainable crop production and is increasingly practiced in different parts of the world. Hence, legume inter-cropping is needed to be promoted under CA to sustain the agricultural systems and secure soil and environmental quality. Legume inter-cropping needs to be promoted under organic farming with proper agro-techniques due to their low input requirements and lower incidence of

insect pests, diseases, and weeds. The unique capability of legume crops of N fixation and recovery of the unavailable form of P in soil is required detailed investigation to minimize the use of expensive N and P fertilizers in future crop production systems. Further, there is a need for advanced breeding and bioinformatics technologies for developing insect-pests resistant, stress-tolerant, and high-yielding varieties to synchronize the food demand and agricultural sustainability (Singh et al., 2021; Sheoran et al., 2021). The development of inter-cropping system-oriented farm machinery and research methodologies is essential to popularize legume inter-cropping among the farmers. Identification of new niches for sustainable intensification and diversification of cereal-based cropping systems through legume inter-cropping is essential for augmenting soil and environmental advantages. The development of supportive policies and markets for regulating the price fluctuations is the need of the hour to make legume inter-cropping more profitable.

16.10 Conclusions

The hallmark quality of legumes makes them an important component of agriculture, agroforestry, and the natural ecosystem in sustaining soil, plant, and environmental health. Legume inclusive inter-cropping systems can alleviate the adverse impact of increasing population pressure, land degradation, ecological damages, and other constraints of crop production through BNF, regulating nutrient cycling, and enhancing the RUE. Legume inter-cropping boosts agronomic yields and profitability along with restoring soil health and environmental quality. Legume-based inter-cropping restores soil fertility by improving the physical, chemical, and biological properties of soil. Legume inter-cropping enriches the soil with essential nutrients (especially N and P), improves SOC content, conserves soil and water, reduces soil erosion and runoff, enhances microbial abundance in the rhizosphere, and mitigates GHGs emissions. The improved soil health ensures feed, food, and nutrient security while advancing ecological sustainability. The inclusion of legume crops not only increases biodiversity but also curbs the extensive use of synthetic N fertilizers. The improved crop diversity, soil biodiversity, and microclimate reduce the incidence of insect pests, diseases, and weeds on crops and thus, maintain plant health and avoid the risks associated with crop failure. Furthermore, legume crops can fit in various cropping systems and thus, are suitable for CA and organic farming systems. Legume inter-cropping contributes to reducing the negative impact of climate variability and securing the livelihood of resource-poor farmers. Thus, legume-based inter-cropping is a vital practice for sustainable crop intensification, food and nutritional security, and soil sustainability.

Abbreviations

BD	Bulk density
BNF	Biological nitrogen fixation
C	Carbon
CA	Conservation agriculture
CEC	Cation exchange capacity
CO ₂	Carbon dioxide
CT	Conventional tillage
DLR	Doubled-up legume rotation
FAO	Food and Agriculture Organization
GHG(s)	Greenhouse gases
K	Potassium
LER	Land equivalent ratio
MBC	Microbial biomass carbon
MBN	Microbial biomass nitrogen
N	Nitrogen
% Ndfa	Percent nitrogen derived from air
N ₂ O	Nitrous oxide
OM	Organic matter
P	Phosphorus
RUE	Resource use efficiency
SDG(s)	Sustainable Development Goals
SOC	Soil organic carbon
SOM	Soil organic matter
SPAD	Soil plant analysis development
UN	United Nations
WUE	Water use efficiency
ZT	Zero tillage

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